Forum:	Disarmament and International Security Committee (GA1)
Issue:	Preventing the use of critical infrastructure sabotage as a means of coercion
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Position:	Co-chair

Personal Introduction

Dear delegates,

My name is Ektoras Moysidis, I am 14 years old and attend the 9th grade in the German School of Athens. It is my utmost honour to serve as one of the Co-chairs in this year's Moraitis Model United Nations, specifically in the Disarmament and International Security Committee. This conference marks my third time chairing and I am sincerely grateful to have been given the opportunity to chair at such a prestigious conference.

Throughout my Model United Nations experience, I have learned many valuable things, such as public speaking, diplomacy and politics, negotiation and much more. This just shows that Model United Nations isn't just a place where you debate complex topics, but also a valuable learning experience that helps delegates develop skills they will carry with them long after the conference ends.

In this guide, you will find a thorough explanation of the topic at hand. However, all delegates are strongly encouraged to engage in research on the topic outside of reading the study guides as it is crucial to have an even wider understanding of the topic that you will be discussing in the committee. I hope the study guides will assist you all in your preparation for the conference and if you encounter any questions or issues regarding the study guide, feel free to contact me via email: ektorasmoysidis@icloud.com. I am really excited to meet you all in October!ektorasmoysidis@icloud.com

Kind regards,

Ektoras Moysidis

Introduction

Role of critical infrastructure sabotage in modern warfare

Critical infrastructure plays a major role in every modern society, providing essential services that people rely on daily. Some examples include energy grids and power lines, water supply systems, public transportation, hospitals and other services that are essential for a country to function. These systems are mostly taken for granted, but their disruption or malfunction can affect almost every part of someone's daily life. Modern societies rely heavily on these systems, meaning that even a temporary disruption can have serious negative effects on both the government and the population. As hybrid warfare continues to advance quickly, these systems become more vulnerable to both physical and cyber attacks.

An attack on critical infrastructure can have consequences reaching beyond the disruption of infrastructure itself. For example, a malfunction or attack on electricity grids and power lines can leave homes, hospitals and businesses without power. Another example, including an attack on transportation systems can prevent people and goods from being able to reach their destinations. Disruptions to water systems can affect access to clean water throughout an entire city, attacks on communication networks can prevent emergency services from communicating. These disruptions in infrastructure can also have major economic losses when businesses or other government services aren't able to operate, since they heavily rely on infrastructure to earn money.

Critical infrastructure as a tool of coercion

Critical infrastructure sabotage can also be used outside of causing economic instability and weakening a government. In some cases, these attacks can be used as a means of coercion in the hope of pressuring the government to change its actions. Instead of solely aiming at destroying infrastructure to cause chaos and instability, the attacker may use the disruption it causes to put pressure on the enemy. This can include forcing a government to change a decision, weaken its support for another country or even make military or economic compromises. This makes critical infrastructure sabotage particularly concerning because its effects can impact civilians who are not directly involved in the conflict. As a result, pressure can build on the government to restore these essential services and take action against the attacker responsible. Critical infrastructure can also be targeted because of its importance to a country's functioning during a conflict. Services such as transportation of goods and military vehicles, natural oil and gas pumps and factories are very important aspects of a country's military. The attacker can then take advantage of such weakened services, the government's weakened political and economic position to prepare or launch an invasion.

Key Terms and Definitions

Critical infrastructure

"The physical structures, facilities, networks and other assets which provide services that are essential to the social and economic functioning of a community or society." [1]

Coercion

"Coercion is the threat or use of punitive measures against states, groups, or individuals to force them to undertake or desist from specified actions." [2]

Hybrid warfare

"The use of a range of different methods to attack an enemy, for example, the spreading of false information, or attacking important computer systems, as well as, or instead of, traditional military action." [3]

NATO's collective defence mandate

NATO's core defense mandate is collective defense, a principle established in Article 5 of the founding 1949 North Atlantic Treaty. Under this mandate, an armed attack against one NATO member state is legally considered an attack against all members, requiring each ally to assist the attacked nation in the manner it deems necessary.

Non-state actor

"A person or an organization that is not part of a government, but is still involved in or important in politics, society, etc. in some way, because of their actions." [4]

Sabotage

"Sabotage is the deliberate destruction of property or slowing down of work with the intention of damaging a business or economic system or weakening a government or nation in a time of national emergency." [5]

Malware

[1] United Nations Office for Disaster Risk Reduction (UNDRR). 2017. The Sendai Framework Terminology on Disaster Risk Reduction. "Critical infrastructure". https://www.undrr.org/terminology/critical-infrastructure.

[2] Khan, M. R. (2015). Coercion | human behaviour. In *Encyclopædia Britannica*. https://www.britannica.com/topic/coercion

[3] Cambridge dictionary. (n.d.). *hybrid warfare*. Dictionary.cambridge.org. https://dictionary.cambridge.org/dictionary/english/hybrid-warfare

[4] Cambridge Dictionary. (2025, February 5). *non-state actor*. @CambridgeWords. https://dictionary.cambridge.org/dictionary/english/non-state-actor

[5] *Sabotage | subversive tactic*. (n.d.). Encyclopedia Britannica. https://www.britannica.com/topic/sabotage-subversive-tactic

"Malware is developed as harmful software that invades or corrupts your computer network. The goal of malware is to cause havoc and steal information or resources for monetary gain or sheer sabotage intent. " [6]

Background Information

Historical use of Infrastructure as a target

Infrastructure has historically been targeted during conflicts because of its importance to a country's military, economy and population. Infrastructure used for transport such as railways, bridges, ports, roads etc., have often been attacked or sabotaged to reduce an opponent's ability to transport troops, military supplies or vehicles. Railways, for example, became increasingly important as armies have depended on them to move troops, weapons and supplies. Therefore, sabotaging railway systems can slow down military operations, and make it way more difficult for an opponent to maintain its forces.

Physical sabotage has also been used as a way of pressuring an opponent, rather than just causing destruction. During conflicts, damaging important infrastructure can create shortages, delays or economic disadvantages, which can greatly affect the ability of a government or army to continue its operations. One major conflict in which this tactic was used was the Second World War, during which military organisations, partisans and other resistance groups carried out sabotage against railway lines, bridges and industrial facilities in order to disrupt military logistics. One of the most well-known examples is railways around Zagreb in Croatia and the Sarajevo-Mostar line in Bosnia and Herzegovina. The resistance was so strong that a resistance group in France, named Résistance Fer („fer" means iron and is part of the French word for railway: „chemin de fer") was created with a principal purpose of disrupting German railways and providing information to other resistance groups.

Infrastructure has also been targeted because its destruction can affect civilians as well as military forces. By sabotaging electricity networks, water facilities, communication systems and factories, the attacker can manage to interrupt the population's daily life while weakening their trust in the government. Aside from having major psychological effects on the population and the government, revolts and protests may arise from the unhappy civilians. In conclusion, this means that by a simple cyber or physical attack an attacker can create fear, economic difficulties, pressure on the government and disrupt essential daily services.

[6] CISCO. (2024). *What is malware? - Definition and examples*. Cisco. https://www.cisco.com/site/us/en/learn/topics/security/what-is-malware.html

Physical Sabotage

Physical sabotage, unlike digital sabotage, requires an attacker to physically disrupt or sabotage infrastructure or use military weapons to damage it. Weapons such as bombs, explosives and air strikes have been used frequently to target bridges, railways, power grids, pipelines and other important facilities. These attacks can greatly weaken the opponent by interrupting transportation, communication, distribution of supplies and essential resources. However, developments in digital technology have introduced another way of targeting key infrastructure. Instead of physically attacking a facility, an attacker may interfere with the computer systems that operate and control it. A very well known example for this phenomenon is Stuxnet malware. In June 2010, Iran's nuclear program was targeted by Stuxnet malware and was sabotaged without the need to use physical attacks. Although digital sabotage has been growing in popularity lately, it does not mean that physical attacks have been completely replaced by cyberattacks, since both methods continue to be used frequently. Instead, cyberattacks have introduced a new way of disrupting infrastructure without requiring an attacker to physically reach the target.

This is particularly important because digital sabotage can be carried out remotely. An attacker operating outside a country may attempt to gain access to information or controls of a power station, transportation system or other facilities without needing to even be present in the country. In April of 2007, Estonia faced one of the largest cyberattacks targeting critical digital infrastructure, which lasted for over 22 days, and happened due to the relocation of a Soviet-era statue in Tallinn. The attacks primarily targeted online services rather than industrial control systems, but rather on government and commercial servers, online banking, public websites and emails. Another famous example where digital and physical sabotage was used, is in Ukraine, namely at the start of the Russo-Ukrainian war in 2022. Ukraine faced cyber attempts to disrupt electricity by operating circuit breakers, while Russian missile strikes on energy infrastructure caused additional widespread power outages.

Physical sabotage can also be carried out in many different ways, not just bombs and nukes. Especially during the Second World War, resistance groups often used explosives, land mines and other methods to damage railway lines, bridges and factories. These attacks were intended to disrupt military operations and logistics, hinder transportation of troops, military vehicles and goods and make it more difficult for occupying forces to control territory.

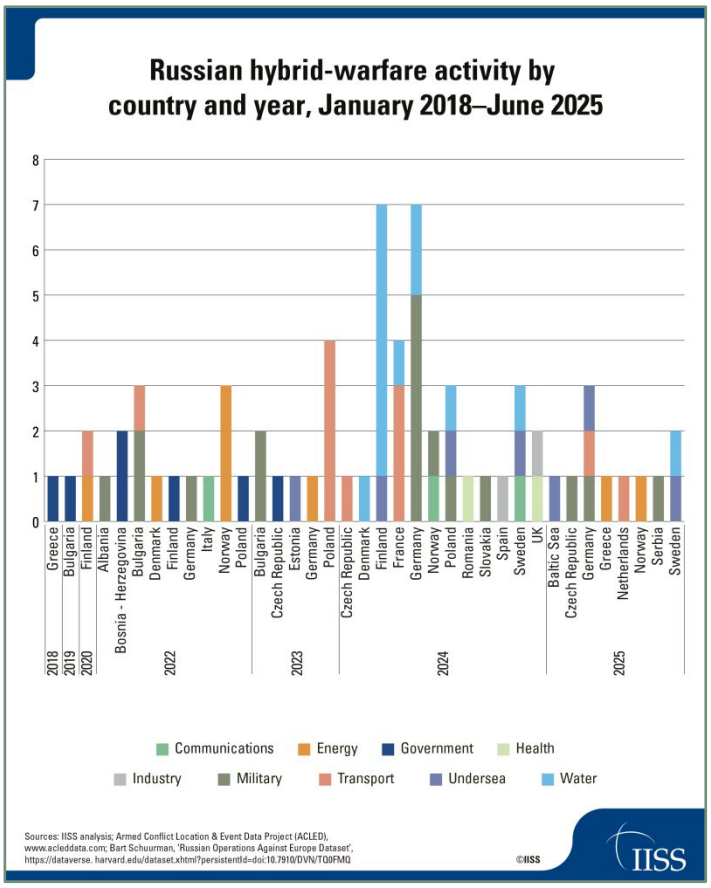


Figure 1: Russian hybrid-warfare activity by country and year, 2018-2025 [IISS][7]

Digital sabotage

Digital sabotage refers to the use of digital systems to disrupt, sabotage or interfere with an operation or a facility. When key infrastructure is connected to computer networks, an attacker may be able to use these systems to affect these facilities physically without being present. Industrial Control Systems (ICS) are particularly important in this context. These systems are used to monitor and control facilities such as power grids, water supplies and transportation. If an attacker gains access to an ICS, they may be able to interfere with its processes, potentially causing equipment to stop working, interrupting services or even causing physical damage. This is one of the many reasons digital sabotage is becoming increasingly common among other methods of sabotage.

Digital sabotage can also be especially relevant to coercion because an attacker does not need to permanently destroy infrastructure to put pressure on a government. Even a temporary disruption to electricity, transportation or communication can have serious effects on civilians, businesses and governments. If the attacker causes this disruption in order to force a government to change its actions, then it is considered a form of coercion. Even when attacks do not have immediate effects on

[7] The scale of Russian sabotage operations against Europe's critical infrastructure. (2019). IISS. https://www.iiss.org/research-paper/2025/08/the-scale-of-russian--sabotage-operations--against-europes-critical--infrastructure/

the government but on the population, this does not mean that the government cannot experience pressure in any form.

Another important difference between physical and digital sabotage is the difficulty of tracking down and identifying the attacker. A physical attack may be detected through visible evidence, such as damaged buildings and infrastructure, from air defence systems, or even from radars. In contrast, a cyberattack can be carried out from compromised computers, servers or accounts which make it difficult to identify the attacker. They may also use systems located in different countries, making it harder to trace the attack to its source. This uncertainty can also make coercion much more difficult to address, since a government will likely not be able to track down the attacker and determine who is responsible. Therefore, without certainty, it will be very difficult for a government to take action against the attacker without risking a response against the wrong organisation or person.

Cyberattacks on Critical Infrastructure

As modern societies have become increasingly dependent on digital systems, cyberattacks against critical infrastructure have become a an important security concern. Power lines, energy grids, hospitals, transportation systems, water networks and communication systems all rely on technology in some way. Sabotaging any of these systems can therefore affect essential services and in many cases cause physical damage.

Ukraine has been a major target of cyber- and physical attacks very frequently since the start of the 21st century. On the 23rd of December 2015, a cyberattack targeted three Ukrainian electricity distribution companies and caused major power outages affecting a quarter of a million people. The attackers used malware to gain access to the companies' systems and interfere with their electricity supply. Another attack which happened a year later in 2016, targeted another Ukrainian electricity transmission company and used malware, similar to the previous attack, to interfere with the control system of an energy grid, which left part of Kyiv temporarily without power. These were among the first attacks of their kind, and demonstrated exactly how cyber sabotage can affect physical infrastructure without needing physical presence. Instead of using bombs or other conventional weapons, attackers were able to interfere with the systems responsible for controlling and distributing electricity. This makes cyberattacks particularly important to the topic of coercion because the distribution of essential services can create enormous pressure on both the civilians as well as the government.

Another example of a cyberattack which inflicted devastating damages, is the colonial pipeline attack in May 2021 in the United States. It was shut down following a ransomware attack, affecting the supply of fuel across parts of the country. Although the attack was financially motivated, it demonstrated how a cyberattack on critical infrastructure can create a wider disruption for the population and economy. An important thing to mention is, that the cyberattack on the United States' Colonial Pipeline was primarily a financially motivated ransomware attack rather than an attempt at political coercion. For it to be considered as a means of coercion it would have to be connected to an attempt to pressure the target into changing a choice or making a particular action.

The use of cyberattacks creates another notable challenge. An attacker can operate from one country, use computer systems in another and target a third country. This emphasises the importance of identifying the attacker when it comes to cyberattacks. Thankfully, the United Nations Convention against Cybercrime, which was adopted by the United Nations General Assembly (UNGA) on 24 December 2024, was created in the hope of strengthening international cooperation in investigating cybercrime and trying to prevent it.

Timeline of Events

Date	Description of Event
April-May 2007	Due to rising political tensions, Estonia faced one of the world's first major cyberattacks targeting critical digital infrastructure. [8]
June 2010	Iran was targeted by Stuxnet malware, specifically targeting its nuclear program and delaying it through technical destruction rather than more common methods such as bombing.
23 December 2015	A cyberattack on three major electric companies in Ukraine caused widespread power outages impacting nearly 225000 people.
7 May 2021	A Russian group of cyber criminals targeted one of the largest oil pipelines in the United States. This pipeline supplied the east coast of the United States with over 45% of the gas and diesel and was forced to shut down. [9]
June 2022	Ukraine was hit with unplanned power outages coinciding with missile strikes on critical infrastructure across Ukraine where four regions were left temporarily without power.
24 December 2024	The United Nations General Assembly (UNGA) adopted the Convention to strengthen international cooperation in preventing and investigating cybercrime. It was adopted by the UNGA in December 2024. [10]

[8] McGuinness, D. (2017, April 27). How a cyber attack transformed Estonia. *BBC News*. https://www.bbc.com/news/39655415

[9] Easterly, J., & Fanning, T. (2023, May 7). *The attack on the colonial pipeline: What we've learned & what we've done over the past two years*. Cybersecurity and Infrastructure Security Agency. https://www.cisa.gov/news-events/news/attack-colonial-pipeline-what-weve-learned-what-weve-done-over-past-two-years

[10] United Nations. (2021b). *United Nations Convention against Cybercrime*. United Nations : Office on Drugs and Crime. https://www.unodc.org/unodc/en/cybercrime/convention/home.html

Major Actors Involved

Russian Federation

The Russian Federation is an important actor in the discussion of critical infrastructure sabotage as a means of coercion because of its use of both cyber and physical attacks carried out in conflict. Russian-linked operations have targeted key infrastructure such as power lines, government services, and other important systems, affecting Ukraine and many more European countries. Meanwhile, Russia has developed its own national policies on protecting important infrastructure from cyberattacks. One of the main laws concerning this issue is Federal Law No. 187-FZ on the Security of the Critical Information Infrastructure of the Russian Federation, which was adopted in 2017 and came into force in 2018. This law establishes specific requirements for protecting critical information infrastructure and aims to ensure that these systems will continue to operate even when subjected to cyberattacks. Furthermore, it establishes responsibilities for organisations which operate information systems in order to improve transparency and enhance the detection, prevention and response to cyberattacks.

Russia has also been involved in several incidents where its cyber operations have been linked to attacks against other countries. Russia and Russian linked groups have also been accused of sabotaging key infrastructure across many European countries within the last decade. Although, Russia has denied responsibility for such attacks. At the same time, the Russian Federation has continued to participate in international discussions about cybersecurity and has supported the development of international law concerning behaviour of states in cyberspace.

Ukraine

Ukraine has been heavily affected by attacks against its critical infrastructure, especially since the beginning of the Russo-Ukrainian war. Ukraine's power systems, government services, communication systems and other key infrastructure have been targeted, leaving its citizens in lack of protection against these cyberattacks, since they do not only affect the government, but the population as well. A well known example includes 23 December 2015, when a major cyberattack on Ukraine's three major electric companies was carried out, and left over 225000 citizens without power. Thankfully, this lasted for only a couple of hours, and didn't cause any long lasting damage. Furthermore, since the beginning of the full scale invasion in 2022, Ukraine has been hit with even more cyberattacks on its infrastructure systems which have significantly affected the country. The effects of these attacks include power cuts, damage to public services and difficulties accessing communication networks. In some cases, these attacks have been planned and carried out at the same time as missile strikes or drone strikes, greatly increasing the disruption caused to the population and the government.

Because of these repeated attacks, Ukraine has introduced several policies aimed at protecting of its key infrastructure. In 2021, Ukraine introduced its cybersecurity strategy, which identified the protection of its infrastructure as one of the country’s main priorities. It focuses on improving

detection and response to cyberattacks, while hoping to increase cooperation between other governments and organisations involved in cybersecurity.

Ukraine has also received significant aid from other countries in order to strengthen its ability to detect, respond and intercept incoming cyberattacks. The North Atlantic Treaty Organisation has also assisted Ukraine economically, as well as with technical support and the exchange of possible threats. Aside from NATO, Ukraine has also received aid from many different countries and organisations as well. This includes nations such as Germany, France, the United Kingdom, the United States, Australia, Japan and many more. Some organisations also include the European Union, by providing financial aid and weapon supplies, the Red Cross which has provided citizens with medical care, food and shelter, and last but definitely not least, the United Nations with relief efforts from its various bodies including the United Nations Refugee Agency (UNHCR), the World Health Organisation and the Ukraine Humanitarian fund which has provided clean water food and healthcare.

Iran

Iran has become an important actor in the field of cyberattacks and infrastructure sabotage, with Iranian-linked groups accused of targeting governments and crucial infrastructure in several countries. One notable example took place on 22 September, when Albania suffered a major cyberattack that disrupted government and public services, as well as websites. NATO and several of its member states traced down the attack on the Government of Iran. This attack disrupted the government's ability to provide essential services to Albania's population and demonstrated how cyberattacks can put pressure on another country without inflicting physical damage.

Iranian cyber groups have developed significantly over the last decade. By gaining access to computer systems, stealing sensitive data, monitoring communications or disrupting, damaging, or disabling services, they can significantly affect a country’s government. These capabilities have become especially important during military conflicts, where cyberattacks can be used alongside military operations to launch an invasion simultaneously while weakening the enemy by interfering with essential services. This is particularly relevant to the ongoing conflict with Israel, where during the conflict, many Iranian linked cyber groups have been accused of targeting the Israeli government’s essential services and other key infrastructure. This can create great pressure, weaken the country and give Iran a chance to weaken the enemy without carrying out a military invasion or physical attack.

Coercion is an especially important part of how cyberattacks and physical infrastructure sabotage can be used during the ongoing conflict. Attacks connected to the war may interfere with the systems that support public services and government operations. This can make it increasingly difficult for authorities to provide services, respond to emergencies and carry out their normal responsibilities. This disruption also pressures decision makers, especially when another country wants to interfere

with their decision making. Therefore, by interfering with these systems, cyber operations can be used to alter a country's decision without requiring a direct physical attack or an invasion.

North Atlantic Treaty Organization (NATO)

NATO plays a critical role in protecting its member states from cyberattacks and threats to critical infrastructure. As modern societies have become increasingly dependent on digital systems, attacks against power grids, communication networks, and other essential services can pose significant security risks. In 2016 NATO formally recognized cyberspace as a domain of operations, alongside land, air and sea. This decision strengthened the alliance's emphasis on cyber defense and encouraged member states to improve their capacity to prevent, detect and respond to cyberattacks.

NATO also promotes greater cooperation among its members in addressing cyber threats. This includes sharing information about potential risks, exchanging knowledge about emerging technologies, and supporting efforts to strengthen national cyber defenses. When a member state experiences a serious cyber incident, for example the 2007 cyberattacks on Estonia, NATO can provide assistance. However, a sufficiently severe cyberattack could also bring Article 5 of the North Atlantic Treaty Organisation in action. Article 5 establishes the principle of collective defence, under which an armed attack against one member is treated as an attack against all members. "The Parties agree that an armed attack against one or more of them in Europe or North America shall be considered an attack against them all and consequently they agree that, if such an armed attack occurs, each of them, in exercise of the right of individual or collective self-defence recognized by Article 51 of the Charter of the United Nations, will assist the Party or Parties so attacked by taking forthwith, individually and in concert with the other Parties, such action as it deems necessary, including the use of armed force, to restore and maintain the security of the North Atlantic area.

Any such armed attack and all measures taken as a result thereof shall immediately be reported to the Security Council. Such measures shall be terminated when the Security Council has taken the measures necessary to restore and maintain international peace and security." [11] Consequently, a cyberattack of sufficient scale and impact could be regarded as an attack on the whole alliance. NATO therefore treats cyber threats as a serious concern and is preparing to respond rapidly and decisively to major incidents.

Nevertheless, the North Atlantic Treaty Organisation's role has limits. Its protection extends only to member states, while many countries affected by attacks on their critical infrastructure are not part of the alliance. Moreover, NATO's primary objective is to defend its members, rather than establish universally applicable rules concerning infrastructure sabotage. This just shows the importance of states to cooperate with the UN in developing principles with broader international application.

[11] *Collective defence and article 5*. (2022). Site Name Seo. https://www.nato.int/en/what-we-do/introduction-to-nato/collective-defence-and-article-5

UN Documents and International Agreements

United Nations Convention against Cybercrime (2024)[12]

"The United Nations Convention against Cybercrime; Strengthening International Cooperation for Combating Certain Crimes Committed by Means of Information and Communications Technology Systems and for the Sharing of Evidence in Electronic Form of Serious Crimes was adopted by the General Assembly of the United Nations on 24 December 2024 in New York, The U.S. by resolution 79/243. The Convention is a global treaty on this matter, which provides States with a range of measures to be undertaken to prevent and combat cybercrime. It also aims to strengthen international cooperation in sharing electronic evidence for serious crimes."

This convention, although it is a significant step towards tackling the issue by improving international cooperation against cybercrime, as it gives nations a common framework for investigating cyberattacks and sharing evidence, has some limitations. Since its main focus is criminal activity, it does not cover the issue as a whole. Furthermore, the Convention is not yet in force; once in force, it will be legally binding on its parties, and its effectiveness will depend on ratification and implementation.

United Nations General Assembly Resolution on Developments in the field of information and telecommunications in the context of international security (2020) [13]

On 31 December 2020, the United Nations General Assembly adopted resolution 75/240, titled "Developments in the field of Information and Telecommunications in the context of International security". The resolution focuses on responsible behaviour by states when using information and communication technologies (ICTs). It supports the idea that states should protect their own key infrastructure from ICT threats and should take steps to prevent harmful cyber activity from being carried out. This resolution establishes expectations of nations when using ICTs and recognises the importance of protecting systems from the ICTs which is a good measure for tackling cyberattacks and coercion as a whole. However, this resolution is not legally binding, therefore its effect is only minimal since its effectiveness also depends on whether or not states choose to follow its principles.

[12] *UN cybercrime convention - full text*. (2021b). United Nations : Office on Drugs and Crime. https://www.unodc.org/unodc/en/cybercrime/convention/text/convention-full-text.html

[13] General, UN. (1999, January 4). *Developments in the field of information and telecommunications in the context of international security :: Resolution /: Adopted by the General Assembly*. UN,. United Nations Digital Library System. https://digitallibrary.un.org/record/265311?ln=en&v=pdf

United Nations Security Council Resolution on the Protection of critical infrastructure against terrorist acts (2017) [14]

The United Nations Security Council Resolution on the Protection of Critical Infrastructure against terrorist attacks was adopted by the Security Council in 2017 and it primarily focuses on protecting critical infrastructure from terrorist attacks. It also encourages cooperation between governments and security authorities, but also private companies operating key infrastructure in the hope of sharing potential threats. This resolution is particularly relevant because it directly addresses the protection of critical infrastructure and encourages international cooperation in order to prevent attacks against it. However, its main focus includes terrorist attacks, which can be an issue since most terrorist attacks are carried out by non-state actors, but it doesn't effectively fix the problem.

Previous Attempts to Solve the issue

Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148

The European Union (EU) NIS2 Directive is the EU's cybersecurity framework which was designed to achieve a high common level of cybersecurity across all member states. It entered into force in January 2023, replacing the previous NIS1 Directive. The NIS2 Directive significantly expanded the scope of the previous framework, now covering 18 critical sectors including energy, transport, banking, health, water supply, digital infrastructure, and public administration.

One of the most important aspects of the Directive is that it is legally binding for all EU member states, meaning that each country was required to incorporate its requirements into national law by October of 2024 . This makes it one of the few cybersecurity frameworks with real enforcement power, as opposed to many international agreements which rely solely on voluntary compliance. The Directive also introduced stronger accountability measures, including the requirement that senior management in organisations take direct responsibility for cybersecurity failures. However, the directive has significant limitations. Since it is a regional framework, it only applies to EU member states, meaning that attacks originating from outside the EU are not directly addressed by it. Moreover, its effectiveness in preventing state-sponsored attacks, which are often responsible for the most damage to infrastructure, is limited as it focuses primarily on improving defences rather than deterring attackers or establishing consequences for those who carry out such attacks.

[14] *Security Council resolution 2341 (2017) [on protection of critical infrastructure against terrorist acts]*. (2017, February 13). United Nations Digital Library System. https://digitallibrary.un.org/record/859472?ln=en

Sanctions against major cyberattack actors

In May 2019, the European Union (EU) established its cyber sanctions regime through Council Regulation 2019/796, which gave it the authority to impose its first ever targeted restrictive measures, such as asset freezes and travel bans, on individuals and entities responsible for cyberattacks which threaten the EU Member States. In July 2020, the EU first imposed cyber sanctions under this framework; in October 2020, it sanctioned individuals and an entity linked to the 2015 cyberattack on the German Bundestag. Similarly the United States has employed a plethora of executive orders, most notably Executive Order 13694 (2015), authorising the blocking of property of persons engaged in significant malicious cyber-enable activities that threaten national security or critical infrastructure. Such examples highlight the degree of accountability that has been created within the cyberspace.

However, these approaches had significant limitations that prevent it from being comprehensive solutions to the issue at hand. Cyber sanctions have designated only a small number of individuals and entities, limiting its overall impact. Moreover, the process of identifying who is responsible for a cyberattack remains highly challenging, as attacks can be routed through multiple countries and systems. Most importantly, sanctions are largely symbolic when applied to state-sponsored actors, as foreign governments and state affiliated groups operating from within their own borders are beyond the reach of asset freezes or travel bans. As a result sanctions serve as a primarily diplomatic tool and a symbolic statement, and action is still required to directly prevent and deter future attacks. Lastly, it should be noted that similar action from the rest of the international community is limited.

NATO's Cyber Defense Policy

The North Atlantic Treaty Organisation (NATO) has been one of the most significant actors in developing collective cyber defence policies aimed at protecting critical infrastructure from sabotage. In July 2016, allies reaffirmed NATO's collective defence mandate and recognised cyberspace as a domain of operations alongside land, air, and sea, marking a turning point in how the alliance approaches cyber threats. This was followed by the adoption of NATO's Cyber Defense Pledge, in which member states committed to strengthening their national cyber defences and improving resilience of critical systems. Furthermore, a sufficiently severe cyberattack against a member state could trigger Article 5 of the North Atlantic Treaty , meaning that cyber operations targeting critical infrastructure could be treated as a hostile act against the entire alliance.

Nonetheless, NATO's cyber defence framework carries important limitations. Primarily, its protections apply exclusively to its 32 member states, leaving the majority of the world's nation-states without the benefit of collective cyber defence. Additionally, NATO's policy focuses on defence and resilience rather than the establishment of a binding international framework that prevents states from using cyberattacks as a tool of coercion. As a result, the need for a broader framework that extends beyond the alliance's membership continues to exist.

Possible Solutions

Strengthening international legal norms

A UN framework addressing the use of critical infrastructure sabotage during conflict could be established. The Disarmament and International Security Committee can use the already existing legal frameworks and the technical expertise of UN agencies to develop principles distinguishing unlawful attacks on civilian infrastructure from lawful operations against military objectives. The framework will be set through the involvement of all UN members to ensure neutrality and respect state sovereignty, creating a universal set of principles that anticipates infrastructure sabotage outside of armed conflict. Although non-binding, the strengthening of international legal norms can in the long term decrease the use of critical installations to achieve a strategic objective. However, bear in mind the mandate of proposing rather than immediately acting the GA1 follows when drafting your proposed resolutions.

Increasing Cooperation

Cooperation between states is essential when addressing cyberattacks against critical infrastructure, particularly because such attacks can originate from one country while targeting another. One possible solution would be the establishment of intergovernmental platforms through which Member States could exchange information about emerging cyber threats, suspicious activities and potential vulnerabilities in their infrastructure. These platforms could operate through existing UN mechanisms and allow national cybersecurity authorities to communicate directly, reducing the time needed to identify threats and coordinate responses.

Furthermore, cooperation between national cybersecurity agencies could be strengthened through joint training programmes, regular meetings and the exchange of technical expertise. Countries with more advanced cybersecurity capabilities could provide assistance to developing states that lack the financial resources or technical knowledge required to protect their infrastructure. This would help improve international preparedness and prevent attackers from exploiting weaker cybersecurity systems. However, such cooperation would require clear rules regarding the confidentiality of sensitive information and respect for national sovereignty, as some governments may be reluctant to share information concerning their critical infrastructure.

Improving Security of Critical Infrastructure

Improving the security of critical infrastructure is another important measure for preventing sabotage and reducing its potential consequences. Governments could invest in stronger cybersecurity systems for essential services, including energy grids, hospitals, transportation networks and water supply facilities. Such investments could involve updating outdated software, strengthening access controls, improving network security and introducing monitoring systems capable of detecting suspicious activity before it causes significant damage.

Another measure would be the introduction of regular cybersecurity inspections and risk assessments for organisations responsible for operating critical infrastructure. These assessments could identify weaknesses in existing systems, evaluate the likelihood of different types of attacks and determine which facilities require additional protection. Governments could establish minimum cybersecurity requirements for operators of essential services, while providing financial and technical assistance to organisations that cannot independently meet these standards.

Finally, developing emergency response plans would allow critical infrastructure operators to maintain essential services or restore them more quickly following an attack. These plans could include backup power supplies, alternative communication systems and procedures for isolating compromised networks. Regular emergency exercises could also help employees and authorities prepare for potential incidents. Although these measures would require substantial investment, they could significantly reduce the vulnerability of essential infrastructure and limit the ability of attackers to use its disruption as a means of coercion.

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